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U–Pb baddeleyite ages linking major Archean dyke swarms to volcanic-rift forming events in the Kaapvaal craton (South Africa), and a precise age for the Bushveld Complex

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ABSTRACT

The Archean basement in the northeastern part of the Kaapvaal craton is intruded by a large number of mafic dykes, defining three major dyke swarms, which collectively appear to fan out from the Bushveld Complex. Herein we present U–Pb baddeleyite ages for two of these dyke swarms, the northwest trending Badplaas Dyke Swarm and the east-west trending Rykoppies Dyke Swarm, and infer their correlation with tectonic events in the Kaapvaal craton. We also present a U–Pb baddeleyite age for a noritic phase of the Marginal Zone of the Rustenburg Layered Suite (Bushveld Complex).

The age of the Badplaas swarm is determined from two dolerites dated at 2965.9 ± 0.7 Ma and 2967.0 ± 1.1 Ma. These ages coincide with units of the Nsuze Group lavas (2967–2985 Ma), which constitute the world's oldest preserved rift basin, and suggest that dykes of this swarm are feeders to basaltic units of this group. Similarly, the E–W trending Rykoppies swarm has earlier been interpreted as a potential feeder system to the Bushveld Complex. However, the emplacement ages of six dolerites fall in the range 2.66–2.68 Ga, thus ~600 Myr earlier than the intrusion of the Bushveld Complex (herein dated at 2057.7 ± 1.6 Ma). Rather, these ages coincide with the Allanridge Formation at the uppermost part of the Ventersdorp Supergroup as well as volcanic rocks of the "protobasinal" sequences preserved at the base of the overlying Transvaal Basin. The Rykoppies Dyke Swarm probably marks the initial stages of rifting of the Transvaal Basin and reflects a major shift from a NW–SE to an E–W trending tectonic setting. The origin of the Rykoppies Dyke Swarm can be linked either to prolonged mantle plume activity or to the onset of back-arc extension associated with south-directed subduction of oceanic lithosphere in a compressional setting along the northern margin of the Kaapvaal craton.

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1. Introduction

The importance of accurate and precise emplacement ages of mafic dyke swarms in combination with paleomagnetic studies is critical for obtaining reliable global plate reconstructions (Halls, 1982; Buchan and Halls, 1990). In deeply eroded cratons dominated by Archean rocks, age determinations of dykes yield invaluable clues to ancient tectonic settings and potentially provide correlations to supracrustal units in the stratigraphic record. Baddeleyite (ZrO_2) is an excellent geochronometer for mafic rocks (Heaman and LeCheminant, 1993) and recent improvements in recovering baddeleyite grains (Söderlund and Johansson, 2002) have significantly enhanced the success rate in dating silica-undersaturated rocks.

The 3.1–3.7 Ga Archean granitoid-greenstone basement in the northeastern part of the Kaapvaal craton, South Africa (Fig. 1; Brandl et al., 2006; Robb et al., 2006) is intruded by numerous mafic dyke swarms with different trends (Fig. 2; Hunter and Reid, 1987). There are three dominant, regional-scale dyke swarms in the Archean basement, which collectively appear to radiate from a centre within the eastern lobe of the ~2.06 Ga Bushveld Complex (Uken and Watkeys, 1997): (1) in the north, a swarm of dense NE-SW trending dykes, which may cross-cut the ~2.66–2.06 Ga Transvaal Supergroup (Eriksson et al., 2006), including the Bushveld Complex (Cawthorn et al., 2006), (2) a centrally positioned swarm of E–W trending dykes, which seems to be restricted to the Archean basement, and (3) in the south, a swarm of NW–SE trending dykes, which cut across the ~3.1–3.6 Ga granitoid-greenstone terrain (Brandl et al., 2006).

In this paper, we present precise U–Pb baddeleyite emplacement ages for the E–W and NW–SE trending pre-Transvaal dyke swarms, referred here to as the Rykoppies and Badplaas Dyke

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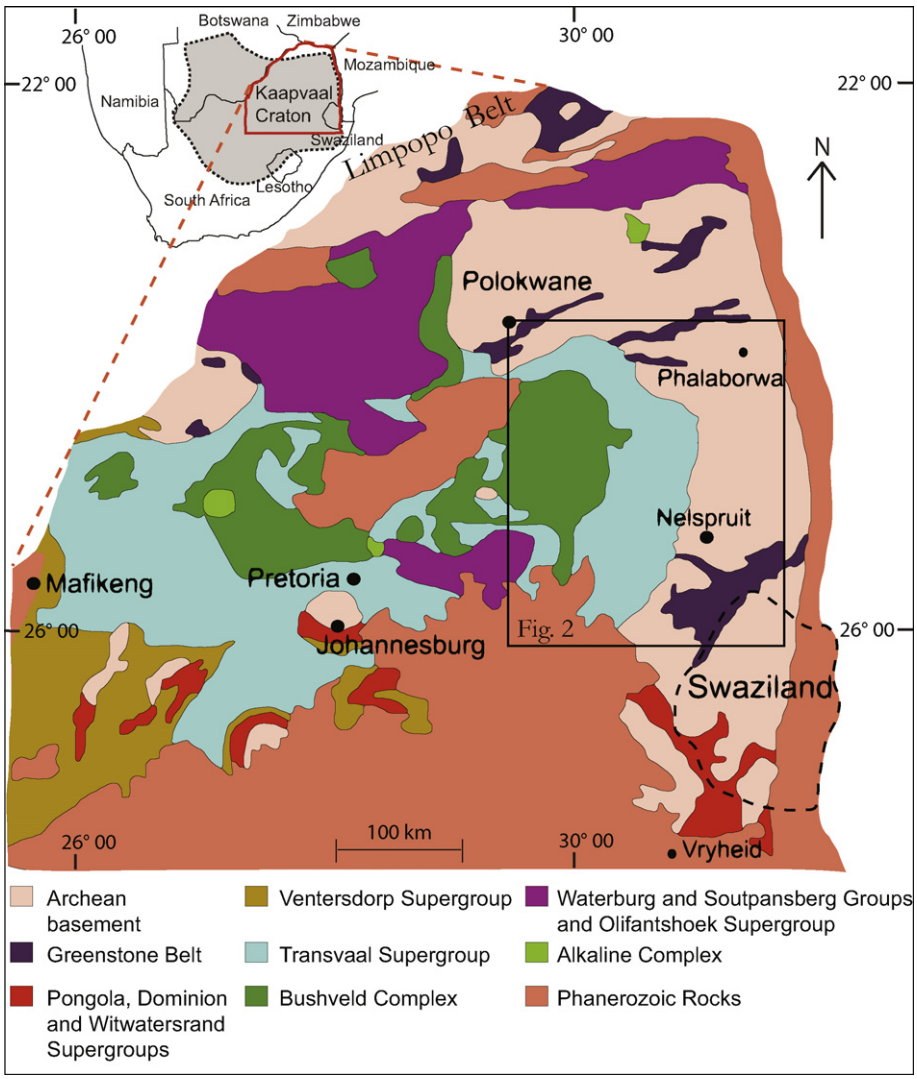


Fig. 1. Simplified geological map of the northeastern part of the Kaapvaal craton, modified from Johnson et al. (2006). The inset image of the outline of the Kaapvaal craton is modified from Poujol et al. (2003).

Swarms, respectively. These results require a re-evaluation of previously proposed linkages to volcano-tectonic events within the relatively well-constrained South African stratigraphy (e.g. Johnson et al., 2006). We also present a precise U–Pb baddeleyite age for the intrusion of the Rustenburg Layered Suite, which contributes to the growing database of precise ages for the Bushveld Complex itself and other magmatic units of similar age across the Kaapvaal craton (e.g. Scoates and Friedman, 2008; Mapeo et al., 2006; Coertze et al., 2006; De Waal et al., 2006; Buick et al., 2001). In contrast the NE-trending dykes are at least in part ca. 1870 Ma (J. Olsson, unpublished U–Pb baddeleyite data). These are not discussed in further detail in this paper.

2. Geology

We first provide a brief overview of the relevant geological and geochronological details for the northeastern part of the Kaapvaal craton, focusing on volcanic sequences (Fig. 1) that are older than 2.0 Ga, as well as the studied Badplaas and Rykoppies Dyke Swarms within the Kaapvaal basement (Fig. 2). For more detailed information, the reader is referred to Hunter and Reid (1987), Hunter and Halls (1992), Uken and Watkeys (1997), Poujol et al. (2003), Eglington and Armstrong (2004), Johnson et al. (2006) and Schmitz et al. (2004).

2.1. The Archean basement and cover sequences

The Ancient Gneiss Complex (AGC) in Swaziland represents the nucleus onto which several granitoid sub-domains amalgamated, along with a number of greenstone belts, and which collectively constitute the Kaapvaal craton (De Wit et al., 1992). The most reliable age-constraints for the AGC are a U–Pb SHRIMP age of 3644 ± 4 Ma on a tonalitic gneiss (Kröner and Compston, 1988) and a U–Pb CA-TIMS (chemical abrasion–thermal ionization mass spectrometry) age of 3662.8 ± 0.5 Ma on a nearby outcrop (Schoene et al., 2008). Zircon xenocrysts within the AGC have yielded older ages of 3702 ± 1 Ma (Pb-Pb evaporation age) and 3683 ± 10 Ma (U–Pb SHRIMP) (Kröner et al., 1996; Kröner and Tegtmeyer, 1994), and may indicate an older component. Among the numerous greenstone belts on the Kaapvaal craton, the 3.5–3.2 Ga Barberton Greenstone Belt is the largest, the oldest and the most rigorously investigated (Kröner and Compston, 1988; Armstrong et al., 1990; Kröner et al., 1996; Kamo and Davis, 1994). U–Pb and Pb–Pb geochronology of different stratigraphic units has supported the

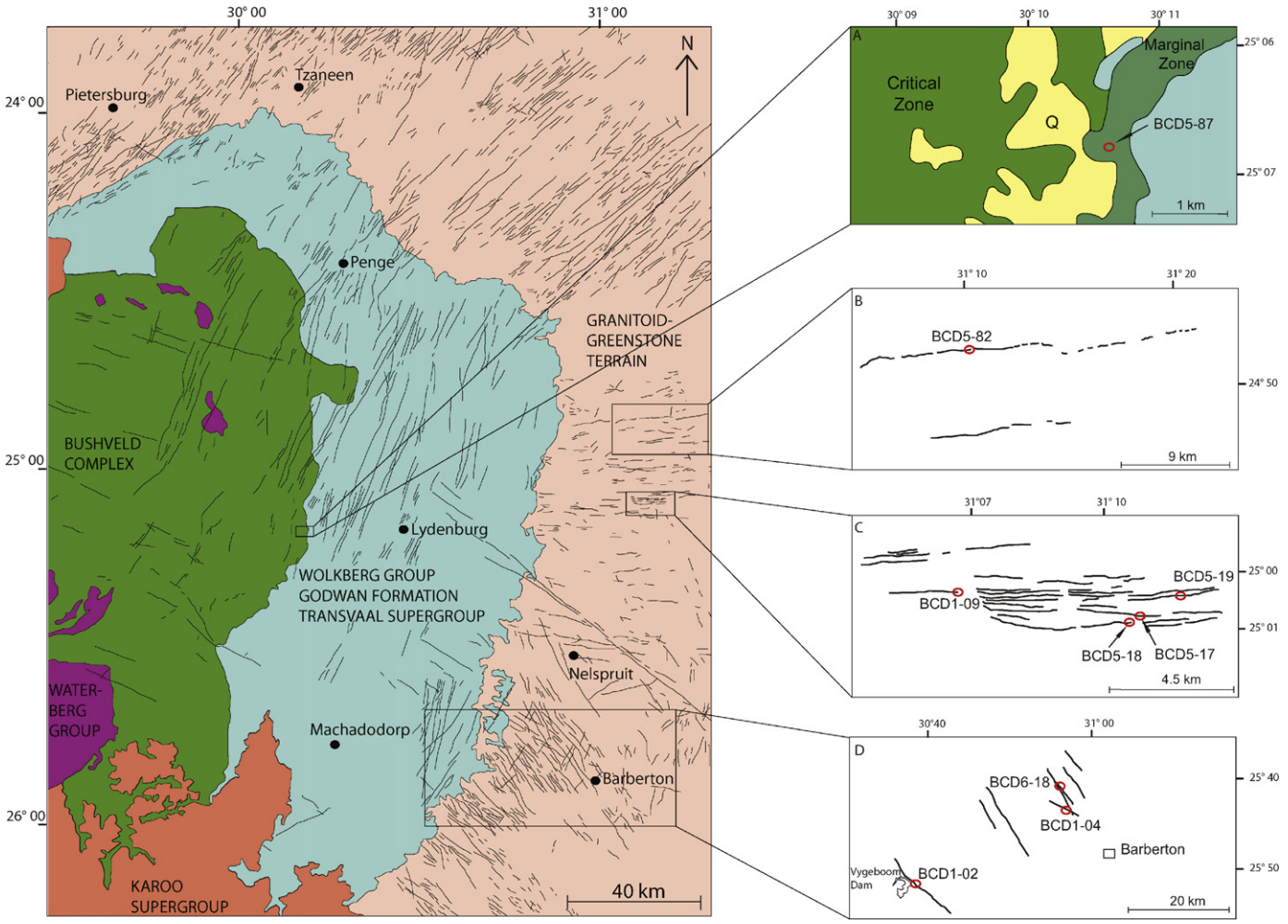


Fig. 2. Map showing the distribution of dolerite dykes in the Archean basement, the Transvaal Supergroup and the Bushveld Complex. The map is modified from Uken and Watkeys (1997) with addition of information from Google Earth. Inset map (A) is modified from Walraven and Hartzer (1986), showing the Critical and Marginal zones of the Bushveld Complex, rocks belonging to the Transvaal Supergroup and quaternary deposits (Q). Inset map (B–D) is based on information from Google Earth.

idea that the Barberton Greenstone Belt is a structural complex consisting of a number of discrete terrains that were amalgamated mainly through tectonic stacking (Kröner et al., 1996). The Barberton Greenstone Belt is surrounded by a number of granitoid batholiths and plutons of varying ages, described in more detail below (see Section 2.2.1).

The 3074 ± 6 Ma Dominion Group (Armstrong et al., 1991) is the oldest preserved cover sequence on the Kaapvaal craton, and it is unconformably overlain by the Witwatersrand/Pongola and Ventersdorp Supergroups. The more easterly occurring Pongola Supergroup correlates with the coeval Witwatersrand Supergroup (Beukes and Cairncross, 1991). Specifically, the 2914 ± 8 Ma Crown Formation lavas near the top of the lower part of the Witwatersrand Supergroup (upper West Rand Group; Robb and Meyer, 1995) and the 2984 ± 2.6 Ma Nsuze Group lava from the lower part of the Pongola Supergroup (Hegner et al., 1993) represent the best evidence for a Witwatersrand–Pongola Supergroup correlation. The age of the Pongola Supergroup is, otherwise, poorly constrained, with an upper age given by the 3107 +4/−2 Ma granitoid basement beneath the Pongola Supergroup (Kamo and Davis, 1994) and a lower age limit given by a cross-cutting granitoid intrusion, dated at 2863 ± 8 Ma (Reimold et al., 1993) and the cross-cutting ~2860 Ma mafic-ultramafic Usushwana Complex (Hunter and Reid, 1987).

A relatively rapid (~35 Myr) deposition of the unconformably overlying volcano-sedimentary sequence of the Ventersdorp Supergroup (Eriksson et al., 2002) is indicated by ages of 2714 ± 8 Ma and 2709 ± 5 Ma for the basal (Klipriviersberg Group) and middle (Makwassie Formation) parts, respectively, of the sequence (Armstrong et al., 1991). Wingate (1998) argues, however, that the significantly older 2782 ± 5 Ma Derderpoort basalts represent the most basal rocks of the Ventersdorp Supergroup and that 2714 ± 8 Ma should be regarded as a minimum age. The Ventersdorp Supergroup appears to have been deposited in a NE–SW trending rift structure (Stanistreet and McCarthy, 1991), and it is debated whether the Ventersdorp flood basalts (mainly Klipriviersberg Group) are related to the collision between the Kaapvaal and the Zimbabwe cratons (Light, 1982; Burke et al., 1985a; Stanistreet and McCarthy, 1991) or to a mantle plume event (Hatton, 1995; Eriksson et al., 2002).

Several groups of volcanic and immature clastic sedimentary rocks unconformably overlie the Ventersdorp Supergroup (Eriksson and Reczko, 1995). These so-called “protobasinal fills” (a purely descriptive term) are exposed as discrete sedimentary basin-fills along the northern and eastern exposed base of the Transvaal Supergroup, and are tentatively interpreted to have been deposited in a rift environment (Eriksson et al., 2001). Two lava ages in the upper Buffelsfontein Group, of 2657–2659 Ma (unpublished report SACS, 1993) and 2664 ± 0.7 Ma (Barton et al., 1995), provide a depositional age for these “protobasinal fill” successions.

The predominantly sedimentary Transvaal Supergroup overlies the “protobasinal” successions within its E–W trending Transvaal Basin, and unconformably overlies the Ventersdorp Supergroup

within the NE–SW trending Griqualand West basin and lesser Kanye basin in Botswana (Eriksson et al., 2001). The best age estimate for the base of the Transvaal Supergroup is derived from the 2642 ± 2.3 Ma Vryburg Formation, in the Griqualand West basin, which is considered to be correlative with the base of the Black Reef Formation in the Transvaal Basin (Walraven and Martini, 1995; SACS, 1980). The 2222 ± 13 Ma Ongeluk-Hekpoort Formation (Cornell et al., 1996) is the youngest dated unit in the Transvaal sequence, but these lavas are conformably overlain by another few thousand meters of sediments.

The top of the Transvaal Supergroup is discordantly overlain and intruded by the Bushveld Complex, which includes (1) an older Rooiberg basalt-rhyolite lava Group (2061 ± 2 Ma; Walraven, 1997), (2) younger Lebowa Granite (2054 ± 2 Ma; Walraven and Hattingh, 1993) and Rashoop Granophyre Suites (2053 ± 12 Ma; Coertze et al., 1978), as well as (3) world's largest (~65,000 km^2) mafic layered intrusion (the Rustenburg Layered Suite), the latter of which hosts the world's richest deposits of platinum group elements, chromium and vanadium (Lee, 1996). Walraven et al. (1990) reported several ages for the Rustenburg Layered Suite at around 2060 Ma. Buick et al. (2001) published a $^{207}Pb/^{206}Pb$ titanite age of 2058.9 ± 0.8 Ma from calcsilicate xenoliths in the Suite's Upper Zone. However, Scoates and Friedman (2008) recently presented a somewhat conflicting zircon Concordia age of 2054.4 ± 1.4 Ma for the Merensky Reef in the lower-lying Critical Zone, arguing that the age by Buick et al. (2001) could be regarded as titanite growth during hydrothermal alteration of the host rock. Connected to the Bushveld Complex are smaller sill-, dyke- or plug-like satellite intrusions in the central parts of the craton (Anhaeusser, 2006; Cawthorn et al., 2006), of which the Moshaneng Complex in Botswana and the Uitkomst Complex in southeastern Kaapvaal, together with the Marble Hall diorite, have been dated by U–Pb geochronology. Zircons and titanites from the Moshaneng granite have been dated at 2054 ± 2 Ma (Mapeo et al., 2004). The basic-ultrabasic layered Uitkomst Complex, which is the host for Ni–Cu–Co–PGE and Au deposits (Gauert et al., 1995), yielded a $^{207}Pb/^{206}Pb$ SHRIMP zircon age of 2044 ± 8 Ma (De Waal et al., 2001), while the age of the chemically similar Marble Hall diorite has been determined to be 2055.6 ± 3.1 Ma by the SHRIMP method (De Waal and Armstrong, 2000).

2.2. Dyke swarms

The Archean basement in the eastern Kaapvaal is intruded by a number of prominent mafic dyke swarms (Fig. 2; cf. Hunter and Reid, 1987; Uken and Watkeys, 1997). Of the many potentially different dyke swarms in the Archean basement, this review is restricted to the NW–SE trending Badplaas Dyke Swarm and the E–W trending Rykoppies Dyke Swarm, for which new ages are reported.

2.2.1. Badplaas Dyke Swarm

The Badplaas Dyke Swarm is ~80 km wide, and can be followed from Swaziland in the southeast for at least 100 km before disappearing under the cover sequences of the Transvaal Supergroup in the northwest (Fig. 2; Hunter and Halls, 1992; Uken and Watkeys, 1997). It has intruded into the AGC, but also into Archean granitoid bodies such as the 3227 ± 4 Kaap Valley pluton (Kamo and Davis, 1994), the 3212 ± 2 Ma Nelshoogte pluton (York et al., 1989) and the 3105 ± 3 Ma Mpuluzi batholiths (Kamo and Davis, 1994). Since the dykes of this swarm are absent in the 2691 ± 2 Ma Mbane pluton (Layer et al., 1989), Hunter and Halls (1992) suggested that the majority of these dykes may have been emplaced between 3.0 Ga and 2.7 Ga.

The dyke swarm parallels the ultramafic ~2.8 Ga Ushushwana Complex (compare Figs. 1 and 2) and previous investigations have associated the swarm with lavas of the ~3.0 Ga Pongola Supergroup (Hunter and Reid, 1987; Uken and Watkeys, 1997). Burke et al. (1985b) stated that rocks of the Pongola Supergroup have been deposited in a parallel, northwest trending rift system. Dykes within the Badplaas swarm more commonly bifurcate towards the northwest, implying a predominant magma movement from the southeast (Hunter and Halls, 1992).

The dolerite dykes can be divided into high-Ti and low-Ti types (Hegner et al., 1993), where the high-Ti dykes have geochemical characteristics that resemble the 2984 ±2.6 Ma lavas from the Nsuze Group. The low-Ti dykes have a geochemical signature that is typical of early Proterozoic dykes (Hunter and Halls, 1992). However, geochemical variations (including that of Ti) can also be explained by variable degrees of crustal contamination and low-pressure pyroxene and plagioclase fractionation of a common primary basaltic melt (Hunter and Halls, 1992). The latter explanation is—with the exception of one high-Ti dyke (BA-1)—more consistent with Klausen et al's (2010) inability to make a similar low-Sr/V and high-Sr/V-discrimination between Hunter and Halls' (1992) high-Ti and low-Ti dykes, respectively.

2.2.2. Rykoppies Dyke Swarm

The E-W trending Rykoppies Dyke Swarm is more than 50 km wide and 100 km long (Fig. 2). The most prominent dyke within this swarm is the Rykoppies dyke (Fig. 2b), which marks the central axis of the dyke swarm and predates the cross-cutting Timbavati Gabbro, of presumed Umkondo age (i.e., ~1.1 Ga; Burger and Walraven, 1979, 1980; Hargraves et al., 1994; Hanson et al., 2004). The Rykoppies Dyke Swarm appears to be exposed only in the Archean basement and is discordantly overlain by the ~2.64 Ga and younger Transvaal Supergroup (Walraven and Martini, 1995), thus leading Hunter and Reid (1987) to propose an Archaean age for the Rykoppies Dyke Swarm. Uken and Watkeys (1997) suggested that the swarm could be a feeder system to the ~2.06 Ga Bushveld Complex, not only because of its east–west trend parallel to the long axis of the complex but also because of a possible link with basement-hosted dykes and sills within the Transvaal Supergroup (e.g. Sharpe, 1981). This view is also upheld in a recent compilation of South Africa's geology (cf., Table 2 in Anhaeusser, 2006).

2.3. Mineral extraction, analytical procedures and data reduction

Baddeleyite was extracted from rock samples weighting 1–2 kg. The extraction was performed at Lund University, Sweden, using the "water-based technique" of Söderlund and Johansson (2002). Approximately 100–150 baddeleyite grains were extracted from each sample and about 10–25 grains were combined in each analysis (see Table 1 for more detailed information).

U–Pb chemistry and mass spectrometry were performed at the Laboratory of Isotope Geology at the Swedish Museum of Natural History in Stockholm. The baddeleyite grains were transferred to Teflon® capsules and washed in 2–3 M HNO_3 on a hot plate and rinsed repeatedly in H_2O. A small amount of a ^{205}Pb–$^{233-236}U$ tracer was added together with a mixture of HF and HNO_3. The baddeleyite grains were completely dissolved after 24 h under high pressure and temperature (~210 °C). After dissolution, samples were dried, re-dissolved in 3.1 M HCl and loaded on 50 μl columns filled with pre-cleaned anion exchange resin (Bio-Rad 200–400 Mesh chloride). The Zr–Hf–REE cut was washed out with 3.1 M HCl and collected in Savillex capsules and saved for future Hf isotope analyses (cf. Söderlund et al., 2006). U and Pb were washed out with H_2O and collected in the same Teflon® capsules as used for dissolution. A small amount of H_3PO_4 was added to each capsule and the samples were put on a hot plate to evaporate overnight. U and Pb were loaded on the same single Re filament together with a small

Table 1
U–Pb baddeleyite data.

Rock unit	Sample	Fraction[a]	U/Th[b]	Pb*/Pbc[c]	$^{206}Pb/^{204}Pb$ meas[d]	$^{207}Pb/^{235}U$ [corr][e]	±2σ (% err)	$^{206}Pb/^{238}U$	±2s (% err)	$^{206}Pb/^{207}Pb$	±2σ (% err)	Corr. coeff.	$^{207}Pb/^{235}U$ [Age, Ma]	$^{206}Pb/^{238}U$	$^{207}Pb/^{206}Pb$	±2σ (Ma)	Discordance (%)
Badplaas Dyke Swarm																	
	BCD1-02	A (8)	14.01	177	8997	17.266	0.175	0.5753	0.166	0.2176	0.052	0.95504	2949.7	2929.8	2963.4	0.8	1.5
		B (15)	5.23	45	2220	13.545	0.152	0.4621	0.143	0.2126	0.049	0.94699	2718.4	2448.9	2925.3	0.8	19.9
		C (9)	10.85	161	8114	16.433	0.269	0.5500	0.263	0.2167	0.053	0.98022	2902.4	2825.4	2956.2	0.9	5.5
		D (10)	11.05	222	11412	16.341	0.347	0.5469	0.340	0.2167	0.069	0.98031	2897.0	2812.3	2956.3	1.1	6.1
	BCD6-18	A (15)	4.76	6	309	16.214	0.278	0.5439	0.139	0.2168	0.087	0.83345	2889.5	2800.0	2952.5	3.3	10.1
		B (12)	6.09	54	2852	15.459	0.319	0.5172	0.306	0.2168	0.116	0.93770	2843.9	2687.3	2956.9	1.4	11.3
		C (10)	7.84	46	2319	16.312	0.218	0.5441	0.150	0.2174	0.137	0.78525	2895.3	2800.8	2961.6	2.2	7.1
		D (10)	4.34	25	1253	16.866	0.201	0.5608	0.187	0.2181	0.070	0.93716	2927.3	2869.8	2967.0	1.1	4.9
		E (15)	8.03	8	439	15.284	0.161	0.5173	0.145	0.2143	0.069	0.90462	2833.1	2687.9	2938.2	1.1	12.7
Rykoppies Dyke Swarm																	
	BCD1-09	A (11)	7.14	151	8177	13.136	0.258	0.5197	0.229	0.1833	0.116	0.89269	2689.5	2697.9	2683.1	1.9	−0.6
		B (15)	5.75	43	2320	12.440	0.337	0.4922	0.268	0.1833	0.183	0.83997	2638.2	2580.6	2682.6	3.0	5.0
		C (9)	3.43	36	1919	12.306	0.330	0.4857	0.231	0.1838	0.209	0.77772	2628.0	2551.9	2687.1	3.5	6.5
		D (13)	6.27	44	2484	11.649	0.368	0.4599	0.311	0.1837	0.183	0.86796	2576.6	2439.3	2686.5	3.0	11.2
		E (10)	5.77	11	577	12.048	0.367	0.4770	0.222	0.1832	0.254	0.73543	2608.1	2514.3	2681.8	4.2	9.3
	BCD5-82	A (16)	13.91	24	1373	12.447	0.520	0.4934	0.427	0.1830	0.258	0.86964	2638.7	2585.4	2679.9	4.3	4.9
		B (12)	13.08	52	2920	12.443	0.534	0.4945	0.514	0.1825	0.126	0.97177	2638.4	2590.0	2675.7	2.1	4.1
		C (19)	10.41	58	3471	11.913	0.558	0.4754	0.543	0.1817	0.127	0.97382	2597.6	2507.4	2668.8	2.1	7.4
		D (12)	8.79	16	885	11.994	0.639	0.4791	0.575	0.1816	0.246	0.92304	2603.9	2523.4	2667.2	4.1	7.4
		E (10)	7.89	27	1588	12.186	0.629	0.4858	0.609	0.1819	0.159	0.96741	2618.9	2552.4	2670.6	2.6	9.0
		F (19)	7.47	35	1884	11.538	0.370	0.4612	0.348	0.1814	0.119	0.94710	2567.7	2444.9	2666.1	2.0	10.4
		G (20)	n.d.	35	1890	11.063	0.338	0.4485	0.313	0.1789	0.120	0.93520	2528.4	2388.4	2642.8	2.0	11.9
	BCD5-17	A (16)	7.09	48	2848	11.690	0.298	0.4683	0.254	0.1810	0.149	0.86598	2579.9	2476.3	2662.3	2.5	8.4
	BCD5-18	A (16)	6.45	57	3223	12.035	0.238	0.4834	0.194	0.1806	0.125	0.85072	2607.1	2541.9	2658.1	1.2	5.4
		B (15)	5.31	85	4675	12.154	0.208	0.4882	0.180	0.1805	0.096	0.81787	2616.4	2563.1	2657.9	1.2	4.4
		C (10)	6.22	27	1434	12.057	0.313	0.4835	0.243	0.1808	0.178	0.82475	2608.8	2542.8	2660.5	2.9	6.1
		D (15)	9.49	13	690	11.630	0.239	0.4679	0.216	0.1802	0.101	0.90653	2575.1	2474.6	2655.2	1.7	9.7
	BCD5-19	A (16)	0.50	53	2274	10.983	0.561	0.4409	0.527	0.1807	0.194	0.93839	2521.7	2354.8	2658.9	3.2	13.6
	BCD1-04	A (18)	5.50	92	5290	12.246	0.276	0.4875	0.261	0.1822	0.111	0.92140	2623.4	2559.8	2672.9	1.8	5.2
		B (15)	5.93	16	875	11.931	0.255	0.4739	0.243	0.1826	0.090	0.93604	2599.0	2500.8	2676.5	1.5	9.0
		C (15)	52.93	9	518	11.059	0.415	0.4436	0.405	0.1808	0.129	0.95099	2528.1	2366.8	2660.3	2.1	14.7
Rustenburg Layered Suite																	
	BCD5-87	A (10)	110.57	252	16014	6.538	0.278	0.3731	0.270	0.1271	0.064	0.97307	2051.0	2044.2	2057.8	1.1	0.8
		B (21)	110.42	236	14292	6.484	0.295	0.3704	0.286	0.1270	0.068	0.97311	2043.7	2031.2	2056.3	1.2	1.4
		C (19)	90.93	381	23159	6.487	0.288	0.3705	0.278	0.1270	0.076	0.96463	2044.1	2031.9	2056.6	1.3	1.4
		D (23)	116.00	494	29961	6.405	0.283	0.3658	0.266	0.1270	0.096	0.94047	2033.0	2009.5	2056.9	1.7	2.7

[a] A–G = fractions, (*x*) = number of crystals/fragments.
[b] U/Th model ratio inferred from $^{208}Pb/^{206}Pb$ ratio and age of sample.
[c] Pb^* = radiogenic lead; Pbc = total common lead (initial + blank lead).
[d] Measured ratio, corrected for fractionation and spike.
[e] Isotopic ratios corrected for fractionation (0.1% per amu for Pb), spike calibration, blank (1–5 pg Pb and 0.2 pg U), and residual initial common Pb.
Initial common Pb corrected with isotopic compositions from the model of Stacey and Kramers (1975) for the age of the sample.

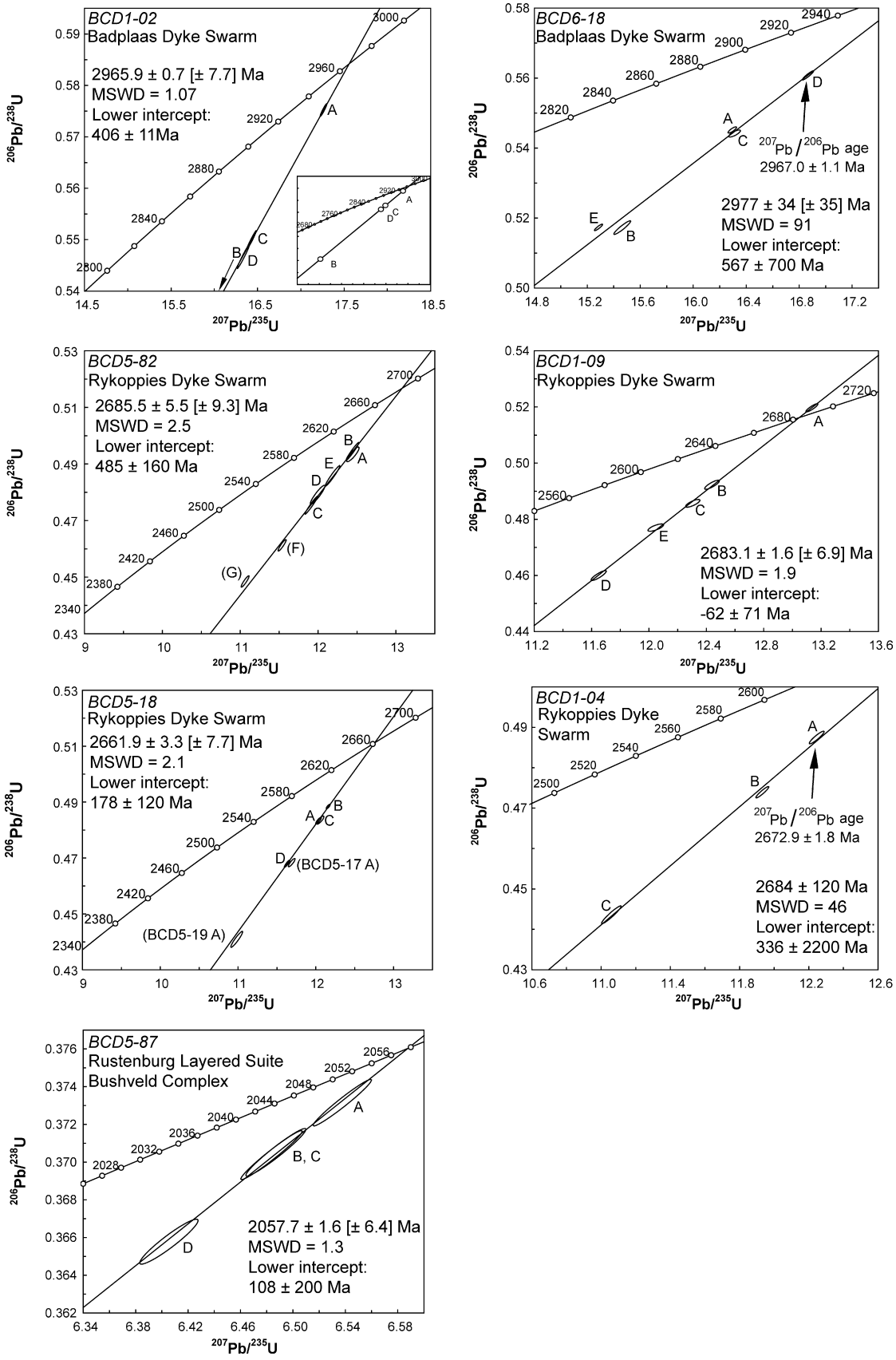


Fig. 3. U–Pb Concordia diagrams showing results from dolerite dykes of the Badplaas Dyke Swarm, Rykoppies Dyke Swarm and a norite from the Rustenburg Layered Suite, Bushveld Complex. Fractions within brackets are excluded from the regressions.

amount of silica gel produced from the recipe of Gerstenberger and Haase (1997).

Mass spectrometry for samples BCD1-02, BCD1-09, BCD5-82 and BCD1-04 (fraction A) [note sample details are given below] were performed on a Finnigan MAT 261 mass spectrometer. The intensities of ^{208}Pb, ^{207}Pb, ^{206}Pb and ^{205}Pb were measured in Faraday collectors. The measurement of ^{207}Pb/^{204}Pb was carried out in dynamic collector mode using an ETP Secondary Electron Multiplier (SEM). BCD5-17, BCD5-18, BCD5-19, BCD5-87, BCD6-18 and BCD1-04 (fractions B-C) were analyzed on a Thermo Finnigan Triton thermal ionisation multicollector mass spectrometer. The intensities of ^{208}Pb, ^{207}Pb, ^{206}Pb and ^{205}Pb were measured using Faraday collectors whereas the intensity of ^{204}Pb was measured simultaneously in an ETP SEM equipped with an RPQ filter. SEM to Faraday gain was controlled by measuring a ~5–10 mV signal between runs. The Pb isotopic measurements were performed at filament temperatures in the 1200–1320 °C range. Isotopes of uranium were measured subsequently in dynamic mode on the SEM at filament temperatures exceeding 1350 °C.

Data reduction was performed using an in-house program written in Microsoft Excel (Per-Olof Persson, Museum of Natural History, Stockholm) with calculations from Ludwig (1991). Analytical results have been calculated and plotted using the Excel Add-in Isoplot (Ludwig, 2003); decay constants for ^{238}U and ^{235}U follow those of Jaffey et al. (1971). All errors in age and isotopic ratios are quoted at the 95% confidence level. Procedural blanks were 1–5 pg Pb and 0.2 pg U and the initial Pb isotope compositions were corrected for using the model compositions of Stacey and Kramers (1975) at 2960 Ma, 2700 Ma and 2060 Ma.

3. Geochronology

Analytical U–Pb data are listed in Table 1 and the results are presented graphically in Fig. 3 (ages reflecting the uncertainties in U decay constants are shown in brackets). Ages of dolerite dykes, their trends and sample locations are summarized in Table 2. Sample sites are shown in Fig. 2A–D. The width and length of the dykes are estimated from outcrops and from Google Earth imagery. These estimates should be regarded as approximate.

Table 2
Sample locations and U–Pb ages.

Rock unit Sample	Location(Lat., Long.)	Dyke trend[a]	Age (Ma)
Badplaas Dyke Swarm			
BCD1-02	25°52′10.81″S 30°39′22.61″E	130°	2965.9 ± 0.7[b] [±7.7 Ma]
BCD6-18	25°40′11.57″S 30°55′44.69″E	140°	2967.0 ± 1.1[c]
Rykoppies Dyke Swarm			
BCD1-09	25° 0′16.67″S 31° 6′55.37″E	90°	2683.1 ± 1.6[b] [±6.9 Ma]
BCD5-82	24°48′19.87″S 31°10′12.00″E	85°	2685.5 ± 5.5[b] [±9.3 Ma]
BCD5-17	25° 0′52.31″S 31°10′45.52″E	90°	2662.3 ± 2.5[c]
BCD5-18	25° 0′59.36″S 31°10′35.36″E	90°	2661.9 ± 3.3[b] [±7.7 Ma]
BCD5-19	25° 0′29.56″S 31°11′32.50″E	90°	2658.9 ± 3.2[c]
BCD1-04	25°42′59.26″S 30°57′1.08″E	115°	2672.9 ± 1.8[c]
Bushveld Igenous Complex			
BCD5-87	25° 6′48.85″S 30°10′37.56″E	–	2057.7 ± 1.6[b] [±6.4 Ma]

[a] Dyke trend estimated from Google Earth in degrees clockwise from true north.
[b] Upper intercept age with decay constant errors in brackets.
[c] ^{207}Pb/^{206}Pb age of a single analysis.

3.1. Badplaas Dyke Swarm

BCD1-02 was sampled in a road-cut close to the Vygeboom Dam north of Badplaas (Fig. 2D), and corresponds to Hunter and Halls' (1992) sample BA-14 (low-Ti group) in their geochemical investigation of the same dyke. This NW-SE trending dyke has a well-exposed strike length of ~15 km, and is up to 100 m wide. Mineral separation resulted in 150 dark-brown, wafer-shaped baddeleyite crystals with typical lengths from 60 μm to 100 μm. Regression through four analyses of BCD1-02 yields an upper intercept age of 2965.9 ± 0.7 Ma (MSWD = 1.1), which is interpreted as the crystallization age of this sample, and a lower intercept of 406 ± 11 Ma (Fig. 3).

BCD6-18 is from a NW-SE trending dyke, northwest of Barberton (Fig. 2D). The length of the dyke is more than 13 km, while the average width is approximately 50 m. It intersects an ENE-WSW trending dyke whose age also is reported herein (BCD1-04; described below). A total of 100 baddeleyite blades and blade fragments, with overall lengths of 30–50 μm, were recovered. The majority of these grains have fine, but dull, overgrowths suspected to be polycrystalline zircon, and only grains with no signs of such rims were selected for isotope analyses. Linear regression involving five fractions from BCD6-18 yields a poorly defined upper intercept age of 2977 ± 34 Ma (Fig. 3), and significant scatter is shown by a high MSWD (91). The scatter among analyses is attributed to trace amounts of polycrystalline zircon rims combined with later disturbance of the U–Pb system. The least discordant analysis (4.9%, fraction D) has a ^{207}Pb/^{206}Pb age of 2967.0 ± 1.1 Ma, i.e. similar to the age of BCD1-02.

3.2. Rykoppies Dyke Swarm

Sample BCD5-82 was collected from the E–W striking Rykoppies Dyke, which has a well-exposed strike length of approximately 40 km and a width between 100–200 m (Fig. 2B). Baddeleyite extraction resulted in approximately 200 dark-brown and wafer-shaped grains with an overall length of 50–100 μm. Regression yields an upper intercept age of 2685.5 ± 5.5 Ma (MSWD = 2.5) and a lower intercept of 485 ± 160 Ma (Fig. 3). The two most discordant analyses (F and G) fall off the Discordia and are omitted from the regression; baddeleyite grains in these fractions may suffer from trace amounts of polycrystalline zircon combined with a complex Pb-loss history.

Sample BCD1-09 was taken from a road-cut north of Hazy View. The dyke has a strike length of 15 km and a width of ~50 m (Fig. 2C). A total of 60 baddeleyite crystals were recovered from this sample. The crystals are dark-brown and typical lengths are 50–100 μm. BCD1-09 yields an upper intercept age of 2683.1 ± 1.6 Ma (MSWD = 1.9) and a lower intercept of −62 ± 71 Ma (Fig. 3).

A major ~N–S trending fault zone separates sample BCD1-09 from more easterly occurring samples BCD5-17, BCD5-18 and BCD5-19, which were all collected from parallel E-W dykes within a relatively restricted area (Fig. 2C). These well-exposed dykes can be followed for ~10 km along strike and are each up to 50 m in width. From BCD5-18, roughly 70 light-brown and small (~50 μm) grains were separated of which the largest and best were selected for U–Pb isotope analyses. From each of samples BCD5-17 and BCD5-19, only ~20 tiny (≤50 μm), light-brown baddeleyite crystals were recovered. The low yield of baddeleyite only allowed for a single U–Pb isotope analysis each for BCD5-17 and BCD5-19.

Regression through four analyses of BCD5-18 results in an upper intercept age of 2661.9 ± 3.3 Ma (MSWD = 2.1), with a lower

intercept at 178 ± 120 Ma. In the Concordia diagram (Fig. 3) both BCD5-17 and BCD5-19 are shown for comparison. The single analyses of these samples are discordant but their $^{207}Pb/^{206}Pb$ ages of 2662.3 ± 2.5 Ma and 2658.9 ± 3.2 Ma, respectively, are indistinguishable from the magmatic, upper intercept age for BCD5-18.

Sample BCD1-04 was collected from an ENE-WSW trending dyke that intersects the BCD6-18 dyke of the Badplaas swarm, northwest of Barberton (Fig. 2D). This dyke has a strike length of 10 km and a width of ~60 m. The trend of this dyke is 25° more south-easterly than the average Rykoppies dykes. Approximately 50 light-brown baddeleyite crystals were recovered during mineral extraction. The grain size was small (~30 μm) and rims suspected to be polycrystalline zircon were observed during optical microscopy. Regression gives a poorly constrained upper intercept age of 2684 ± 120 Ma and a high degree of scattering of the data (MSWD = 46) (Fig. 3). The least discordant fraction (5.2%, fraction A) gives a $^{207}Pb/^{206}Pb$ age of 2672.9 ± 1.8 Ma.

3.3. Bushveld Complex (Rustenburg Layered Suite)

Sample BCD5-87 represents a coarse grained norite collected from an outcrop occurring ~30 km to the west of Lydenburg (Fig. 2A). The Marginal Zone of the Bushveld Complex is ~150 m thick (locally up to 800 m thick), and consists of an unlayered but heterogeneous zone of norites. The Marginal Zone only occurs along the base of the intrusion, on-lapped by a subsequently deposited cumulate stratigraphy (Cawthorn et al., 2006). BCD5-87 is located within a unit mapped as norite/pyroxenite which constitutes the Marginal Zone of the eastern limb of the Bushveld Complex (Walraven and Hartzer, 1986). Baddeleyite extraction yielded more than 200 large (~100 μm), dark-brown and wafer-shaped baddeleyite crystals. Regression of four fractions yields a well defined upper intercept age of 2057.7 ± 1.6 Ma (MSWD = 1.3) with a lower intercept at 108 ± 200 Ma (Fig. 3).

4. Discussion

4.1. Badplaas Dyke Swarm

4.1.1. Correlation with the Nsuze Group lavas

Uken and Watkeys (1997) considered the NW-trending Badplaas Dyke Swarm to be associated with rifting at ~3.0 Ga, linked to the volcanic units of the Dominion and Nsuze Groups. However, their interpretation was based on the observation that the Badplaas swarm structurally predates a "protobasinal" sequence at the base of the Transvaal Supergroup (Godwan Formation; Visser, 1956). Furthermore, Hunter and Halls (1992) noted that many NW-trending dykes cut ~3.0 Ga-old tabular granitoids but are not abundant in the ~2.7 Ga-old Mbabane Granite; providing a maximum and minimum age for a majority of NW-trending dykes. The ages of the two Badplaas dykes determined in this paper (2965.9 ± 0.7 Ma and 2967.0 ± 1.1 Ma; Table 2), lie within these estimated age-constraints and are more precisely synchronous with a 2968 ± 6 Ma porphyritic rhyolite, near the top of the Nsuze lavas (Griffin, 2002).

In the northern exposed parts of the Pongola Supergroup (Hartland area), the 4.6 km thick Nsuze Group is made up of six formations, dominated by volcanic rocks with subordinate sedimentary beds. These rocks were deposited within a period of approximately 15 million years, bracketed between the age of a lower 2985 ± 1 Ma and 2984 ± 2.6 Ma unit (Hegner et al., 1994, 1993) and an upper 2968 ± 6 Ma unit (Griffin, 2002). The BCD1-02 dyke was characterized geochemically by Hunter and Halls (1992; sample BA-14 in their study) as belonging to a group of low-Ti dykes, which in contrast to high-Ti dykes, were inferred to be early Proterozoic in age. The new age result of 2965.9 ± 0.7 Ma presented here for BCD1-02 suggests, however, that the low-Ti dykes are age equivalents to the Nsuze lavas—that is, both groups can be linked to the same Pongola rifting event. This interpretation can tentatively be extended to encompass all the NW–SE trending dykes in this part of the craton, which furthermore appear to converge onto the present exposure of Nsuze lavas in the SE part of the craton, an interpretation also supported by geochemical correlations (Klausen et al., 2010). However, the complexity of this area is exemplified by the BCD1-04 dyke whose age instead indicates a connection to the 2685–2660 Ma Rykoppies Dyke Swarm.

4.2. Rykoppies Dyke Swarm

4.2.1. Correlation with the Ventersdorp–Transvaal Supergroup boundary

The scarcity of geochronological data from the upper Ventersdorp and lower Transvaal Supergroups make correlations with our dyke ages difficult. One possible correlation of the older (~2683–2686 Ma) Rykoppies dykes (BCD1-09 and BCD5-82) is to the basaltic andesite lavas of the Allanridge Formation, since this uppermost part of the Ventersdorp Supergroup is younger than its 2709 ± 5 Ma middle part (Makwassie Formation; Armstrong et al., 1991). This inference agrees with a similarity in geochemical characteristics between the Rykoppies dykes and lavas from the Allanridge Formation (Klausen et al., 2010). The age of BCD5-18 (2661.9 ± 3.3 Ma) dates the emplacement of a younger sub-swarm within the Rykoppies Dyke Swarm. The single analyses of BCD5-17 and BCD5-19 fall along the regression line of BCD5-18 and these dykes are interpreted at present to be of the same generation. Despite significant scatter among analyses of BCD1-04, the $^{207}Pb/^{206}Pb$ age (2672.9 ± 1.8 Ma) of the least discordant fraction places this dyke within the range of 2660–2685 Ma. The later intrusive stage of the Rykoppies swarm coincides with known ages for the "protobasinal" sequences (2657–2659 Ma and 2664 Ma for felsic lavas in the Buffelsfontein Group; unpublished report SACS, 1993; Barton et al., 1995), at the base of the Transvaal Supergroup.

4.2.2. The onset of rifting of the Transvaal Basin

Our results suggest that at 2.66–2.68 Ga, the northeastern Kaapvaal craton was subjected to extension in an approximate N–S direction, which allowed mantle-derived magmas to be emplaced as E–W trending dykes. The timing of the Rykoppies swarm appears to coincide with a shift from mainly volcanic rocks of the Ventersdorp Supergroup to predominately sedimentary rocks of the Transvaal Supergroup. According to Burke et al. (1985a) and Van der Westhuizen et al. (1991), the Allanridge Formation was formed during renewed rifting of the Kaapvaal Craton. The rift structures of the Ventersdorp Supergroup have been assigned a NE-trend (Stanistreet and McCarthy, 1991), but seismic investigations also reveal the presence of E-W trending faults within rock units of the Venterdorp Supergroup (Tinker et al., 2002). Eriksson et al. (2001) proposed that the "protobasinal" successions at the base of the Transvaal Supergroup were deposited within a rift type extensional tectonic setting and infer a tectonic model, where the "protobasinal" successions accumulated in ENE–WSW trending rift systems. The age correlations of the younger sub-swarm of dykes with the "protobasinal" sequences and the proposed correlation between the older sub-swarm and the Allanridge Formation support the idea that the 2.66–2.68 Ga Rykoppies Dyke Swarm marks the onset of rifting of the Transvaal Basin, at a time when the rift system also changed direction from NE–SW to becoming E–W trending.

4.2.3. Tectonic models

Hatton (1995) and Eriksson et al. (2002) regard the Ventersdorp Supergroup as the product of a mantle plume, possibly

related to a global mantle plume event at ~2.7 Ga (Condie, 1998, 2001). Hatton (1995) states that the Ventersdorp magmas were extracted from the mantle at shallow depths (~18–40 km) and subsequently underwent crustal contamination. A fast lava extrusion rate up through the lower half of the Ventersdorp Supergroup (2714–2709 Ma; Armstrong et al., 1991) is consistent with typical flood basalt provinces. If correct, the Rykoppies Dyke Swarm might be a product of prolonged plume activity, connected to the younger eruption of the Allanridge Formation. A mantle plume origin would agree with a slightly radiating pattern of the Rykoppies swarm (here defined by the NW-trending BCD1-04 dyke, and the E–W trend of the main part of the swarm).

In contrast to the mantle plume model, Burke et al. (1985a) and Stanistreet and McCarthy (1991) proposed a rifting model related to the collision between the Kaapvaal and Zimbabwe cratons at ~2.7 Ga, where the Rykoppies Dyke Swarm can be envisaged as emplaced within a hinterland graben setting. The 2671 ± 2 Ma Matok Granite in the Southern Marginal Zone of the Limpopo Belt (Barton and van Reenen, 1992) can be temporally linked to the Rykoppies swarm, inferring synchronous emplacement of mantle-derived magmas in the eastern Kaapvaal craton and granite intrusions further to the north. Furthermore, peak metamorphism of the Southern Marginal Zone has been constrained to 2691 ± 7 Ma (Kreissig et al., 2001). However, the timing of the Kaapvaal–Zimbabwe collision is debatable, because petrological and chronological studies show that metamorphism and substantial transpressional shear within the Central Zone of the Limpopo Belt occurred much later, at approximately 2.0 Ga (Kamber et al., 1995; Buick et al., 2006; Van Reenen et al., 2008; Kröner et al., 1998; Jaeckel et al., 1997). There are, furthermore, no major dyke swarms and LIPs older than 2.0 Ga that mutually exist in both the Kaapvaal and the Zimbabwe cratons, supporting the notion that an amalgamating collision forming the Kalahari craton must have post-dated 2.0 Ga (cf. Söderlund et al., 2010).

The proposed late formation of Kalahari does not rule out a continental back-arc setting as a plausible tectonic model for the Rykoppies Dyke Swarm at c. 2.7 Ga. The Rykoppies swarm could have developed in a back-arc rift system at the same time as the northern margin of the Kaapvaal craton underwent compressional tectonics during south-directed subduction of oceanic lithosphere. Support for this model lies in the interpretation that the "protobasinal" sequences at the base of the Transvaal Supergroup represent intracratonic depositories, whose provenance can be linked to erosion of Limpopo Belt sedimentary source rocks (Eriksson et al., 2002 and references therein).

4.3. Bushveld Complex

4.3.1. Intrusion age of the Bushveld Complex

The upper intercept age of 2057.7 ± 1.6 Ma for the Marginal Zone in the Rustenburg Layered Suite is the first published U–Pb baddeleyite age for the Bushveld Complex. According to the compilation of geochronological data by Scoates and Friedman (2008), the duration of Bushveld-related magmatism occurred over a period of 5 Myr, whereas Cawthorn and Walraven (1998) suggested that the entire Rustenburg Layered Suite was emplaced in only 75,000 years or less. Their calculation was based on thermal modeling of emplacement of a series of successive injections of magma closely spaced in time so that the previous magma had not cooled and differentiated significantly before the next was emplaced. The U–Pb result of the 2057.7 ± 1.6 Ma Marginal Zone (this study) does not solve the problem of the duration of magma emplacement, but contributes to the timing of initial emplacement of the Rustenburg Layered Suite.

5. Conclusions

We report the intrusive ages of two major dolerite dyke swarms that are extensively exposed in the Archean basement in the northeastern Kaapvaal. These results can be linked to distinct Archean magmatic events, when the Kaapvaal craton must have experienced extension (rifting), volcanism and associated mafic dyke intrusions.

Baddeleyite U–Pb analyses of two dolerite dykes from the NW–SE trending Badplaas Dyke Swarm yield similar emplacement ages of 2965.9 ± 0.7 Ma (upper intercept age) and a 2967.0 ± 1.1 Ma (model $^{207}Pb/^{206}Pb$ age), which confirm previous suggestions (Hunter and Halls, 1992; Uken and Watkeys, 1997) for a correlation with coeval basaltic-rhyolitic Nsuze lavas (Hegner et al., 1993; Hegner et al., 1994; Griffin, 2002), within the Pongola Supergroup. This confirms that the NW-bifurcating dykes within the Badplaas swarm were emplaced within one of the earliest known continental rifts (Burke et al., 1985b), originating from a mantle source region that probably underlay the more south-easterly occurring Pongola outcrops.

Baddeleyite U–Pb analyses from six dolerite dykes define a 2685–2660 Ma age-range within the E–W trending Rykoppies Dyke Swarm, which overlaps the major, unconformable transition from a predominantly volcanic 2.78–<2.71 Ga Ventersdorp Supergroup to a predominantly sedimentary 2.64–2.06 Ga Transvaal Supergroup (Armstrong et al., 1991; Wingate, 1998; Walraven & Martini, 1995; Eriksson et al., 2006). In greater detail, two significantly older Rykoppies dykes (2683–2685 Ma) may be correlated with the youngest known lavas in the uppermost part of the Ventersdorp Supergroup (i.e., Allanridge Formation) while three significantly younger Rykoppies dykes (2659–2662 Ma) are coeval with oldest known lavas in the lowermost, "protobasinal", part of the Transvaal Supergroup (i.e., Buffelsfontein Group; unpublished report SACS, 1993; Barton et al., 1995). Our results thereby imply that Ventersdorp volcanism and associated NE–SW orientated rifting ended during a major tectonic shift where E–W trending and volcanic "protobasins" were established as precursors to the prolonged deposition of the Transvaal Supergroup. While this tectonic shift also coincided with emplacements of older granitoid bodies within the Southern Marginal Zone of the Limpopo Belt, the emplacement of the Rykoppies swarm is not likely to be linked with a contemporaneous collision between the Kaapvaal and Zimbabwe cratons as new evidence points towards a much later collision, at ~2.0 Ga. However, it is still possible that the Rykoppies Dyke Swarm formed within a continental back-arc setting, as the northern margin of the Kaapvaal craton (present day coordinates) was a subduction zone. Another possibility is prolonged (or multi-pulsed) plume activity that has been proposed as an explanation for flood basalts within the Ventersdorp Supergroup (Hatton, 1995; Eriksson et al., 2002); supported here by a possible radiating pattern around the eastern limb of the Bushveld Complex.

Even if our results rule out any feeder relationship between the 2057.7 ± 1.6 Ma Bushveld (this study) and the c. 600 Myr older Rykoppies Dyke Swarm (Uken and Watkeys, 1997; Anhaeusser, 2006), the conspicuously E–W elongated geometry of this enormous layered mafic intrusion may, somehow, still have been controlled by those pre-existing rift structures that were apparently established during the emplacement of the Rykoppies Dyke Swarm.

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